



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Course: Fundamentals of Data Science

Course Code: DSA0404

Slot: A

Faculty: KRISHNARAJ M

Submitted By

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Data Manipulation using Pandas in Python

1. Handling Missing Data

Code:

```
*handlingdata.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/handlingdata.py (3.12.3)*
File Edit Format Run Options Window Help

import pandas as pd
import numpy as np

# Sample data with missing entries
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'Age': [25, np.nan, 30, 22],
    'Salary': [50000, 60000, np.nan, 45000]
}

df = pd.DataFrame(data)

# Check for missing values
print("Missing Count:")
print(df.isnull().sum())

# Fill missing Age with median
df['Age'] = df['Age'].fillna(df['Age'].median())

# Drop rows where Salary is missing
df = df.dropna(subset=['Salary'])

print("\nCleaned DataFrame:")
print(df)
```

Output:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/handlingdata.py
Missing Count:
Name      0
Age        1
Salary     1
dtype: int64

Cleaned DataFrame:
   Name  Age  Salary
0  Alice  25.0  50000.0
1   Bob   25.0  60000.0
3  David  22.0  45000.0
>>>
```

2. Filtering Rows Based on Multiple Conditions

Code:

```
filteringrows.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/filteringrows.py (3.12.3)
File Edit Format Run Options Window Help
import pandas as pd

data = {
    'Employee': ['Amy', 'John', 'Raj', 'Kevin', 'Sora'],
    'Department': ['HR', 'IT', 'IT', 'Sales', 'IT'],
    'Experience': [3, 7, 1, 5, 6]
}

df = pd.DataFrame(data)

# Filter employees in IT with experience greater than 5 years
filtered_df = df[(df['Department'] == 'IT') & (df['Experience'] > 5)]

print(filtered_df)
```

Output:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/filteringrows.py
Employee Department Experience
1 John IT 7
4 Sora IT 6
>>>|
```

3. Aggregating and Grouping Data

Code:

```
Groupingdata.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/Groupingdata.py (3.12.3)
File Edit Format Run Options Window Help
import pandas as pd

data = {
    'Store': ['North', 'South', 'North', 'West', 'South', 'West'],
    'Sales': [1200, 1500, 900, 1100, 1700, 1300],
    'Employees': [5, 7, 5, 4, 8, 4]
}

df = pd.DataFrame(data)

summary = df.groupby('Store').agg({
    'Sales': 'sum',
    'Employees': 'mean'
}).reset_index()

print(summary)
```

Output:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/Groupingdata.py
Store Sales Employees
0 North 2100 5.0
1 South 3200 7.5
2 West 2400 4.0
>>>|
```

4. Python Program Using describe ()

Code:

usingdescibe.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/usingdescibe.py (3.12.3)

```
File Edit Format Run Options Window Help
import pandas as pd

data = {
    'Age': [25, 30, 35, 40, 45],
    'Salary': [50000, 60000, 75000, 90000, 110000],
    'Department': ['HR', 'IT', 'IT', 'Marketing', 'HR']
}

df = pd.DataFrame(data)

print("----- Numeric Summary -----")
print(df.describe())

print("\n----- Categorical Summary -----")
print(df.describe(include='str'))
```

Output:

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

```
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/usingdescibe.py
----- Numeric Summary -----
               Age      Salary
count  5.000000    5.000000
mean   35.000000   77000.000000
std     7.905694   23874.672773
min    25.000000   50000.000000
25%    30.000000   60000.000000
50%    35.000000   75000.000000
75%    40.000000   90000.000000
max    45.000000  110000.000000

----- Categorical Summary -----
      Department
count          5
unique         3
top           HR
freq           2
>>>
```

5.head() and tail()in Python Code:

head&tail.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/head&tail.py (3.12.3)

File Edit Format Run Options Window Help

```
import pandas as pd

data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Eva', 'Frank', 'Grace'],
    'Score': [85, 92, 78, 95, 88, 72, 91]
}

df = pd.DataFrame(data)

print("Top 5 Rows")
print(df.head())

print("\nTop 2 Rows")
print(df.head(2))

print("\nBottom 3 Rows")
print(df.tail(3))
```

Output:

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/head&tail.py
Top 5 Rows
      Name  Score
0   Alice     85
1    Bob     92
2  Charlie     78
3   David     95
4    Eva     88

Top 2 Rows
      Name  Score
0   Alice     85
1    Bob     92

Bottom 3 Rows
      Name  Score
4    Eva     88
5  Frank     72
6  Grace     91
>>> |
```